

# The Dissolved Helix and Its Orthogonal Partner

Terrain, rotation, and the completion of  $\mathbb{Z}/4\mathbb{Z}$

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## Abstract

The self-action of the golden companion  $R$  ( $R^2 = R + I$ ) is a single real-axis mode: its commutator spectrum is  $\{0, \pm\sqrt{5}\}$  and its angle charge lies in the sublattice  $\{0, 2\} \subset \mathbb{Z}/4\mathbb{Z}$ . We show, entirely from the exact structure, that this mode cannot *couple to a copy of itself* into a bound state — the generative grade  $[R, R]$  vanishes, self-composition keeps the charge in  $\{0, 2\}$  while the Mahler measure grows without ever returning to the cyclotomic floor, and the “on-circle-and-expanding” (Salem) channel is empty. Closure requires the orthogonal charges  $\{1, 3\}$ , and these are supplied by exactly one seed: the quartic  $K = x^4 + 5x^2 - 5$ , whose real roots  $\pm K$  are the *terrain* (in the field  $\mathbb{Q}(5^{1/4}) \supset \mathbb{Q}(\sqrt{5})$ ) and whose imaginary roots  $\pm i\beta$  are a  $\frac{\pi}{2}$  *rotation about the origin*. The discriminant flip  $D = 1 + 4C$  makes terrain and rotation the two faces of one event; the golden strand is the  $D > 0$  face and  $K$  straddles the fold, re-adjointing the  $D < 0$  face. Every result carries an epistemic tag and, where it exists, the repository file that checks it; all displayed constants are machine-verified in exact arithmetic. The “helix” reading is a **[POSITED]** overlay on a **[FORCED]** substrate — but the terrain/rotation anatomy it rests on is the repository’s own. Finally we abstract the procedure that produced these results: a *harness* that imports an external structure, adjudicates each of its claims against the exact substrate, and partitions them into survivors, casualties, and residue (§8). The helix is its first worked instance; the transferable object is the harness, not the helix.

## 1 Epistemic conventions

Claims are tagged. **[FORCED]**: a theorem of the exact structure, machine-checked in the pipelines. **[COMPUTED]**: an exact fact derived here (symbolic, no floating point crosses a decision). **[ESTABLISHED]**: standard external mathematics. **[POSITED]**: an interpretive overlay (the “helix” language), grounded but not load-bearing. **[OPEN]**: acknowledged open. *No float crosses a decision boundary*: every constant below is decided over  $\mathbb{Q}$ ,  $\mathbb{Q}(\sqrt{5})$ , or  $\mathbb{Q}(5^{1/4})$ . Appendix A lists the checks. A claim the substrate *falsifies* (its negation forced by an exact counterexample) is marked **[REFUTED]**; equivalently **[REFUTED]**  $\gamma \equiv$  **[FORCED]**  $\neg\gamma$ .

## 2 The keystone: the golden generator

**Definition 2.1** (Golden data). Let  $\varphi = \frac{1+\sqrt{5}}{2}$  and  $\psi = \frac{1-\sqrt{5}}{2} = 1 - \varphi = -\varphi^{-1}$ , the roots of  $x^2 - x - 1$ . Let  $R = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$  be the Fibonacci companion.

**Proposition 2.2** (The keystone).  $\text{charpoly}(R) = \lambda^2 - \lambda - 1$ , so  $\text{Tr } R = 1$ ,  $\det R = -1$ ,  $\text{spec}(R) = \{\varphi, \psi\}$ , and  $R^2 = R + I$ . Moreover  $\varphi\psi = -1$ ,  $\varphi + \psi = 1$ , and

$$\varphi - \psi = \sqrt{5} = \varphi + \varphi^{-1}. \quad (1)$$

**[FORCED]** (drop-one keystone appendix of the  $\lambda = 2c$  paper; `test_p1_05_keystone.py`).

*Proof.*  $\det(\lambda I - R) = \lambda(\lambda - 1) - 1 = \lambda^2 - \lambda - 1$ , whose roots are  $\varphi, \psi$ ; Vieta gives  $\varphi + \psi = 1$ ,  $\varphi\psi = -1$ . Cayley–Hamilton yields  $R^2 - R - I = 0$ . Finally  $\varphi - \psi = \frac{(1+\sqrt{5})-(1-\sqrt{5})}{2} = \sqrt{5}$ , and  $\varphi + \varphi^{-1} = \varphi + (\varphi - 1) = 2\varphi - 1 = \sqrt{5}$ .  $\square$

**Proposition 2.3** (The self-action). *Let  $\text{ad}_R = [R, \cdot]$  act on  $M_2(\mathbb{R}) \cong \mathbb{R}^4$ . Then*

$$\text{spec}(\text{ad}_R) = \{0, 0, +\sqrt{5}, -\sqrt{5}\} = \{0, \pm\sqrt{5}\}, \quad (2)$$

and  $\sqrt{5}$  (the sole irrational in the spectrum) is the coupling  $\varphi + \varphi^{-1} = \varphi - \psi$ . [FORCED] (*test\_p2\_07\_uniform.py, Lem. 8.1*).

*Proof.* For a diagonalisable  $R$  with eigenvalues  $\mu_1, \mu_2$ , the operator  $[R, \cdot]$  has eigenvalues  $\{\mu_i - \mu_j\}_{i,j}$  on  $M_2$ . Here  $\{\varphi - \varphi, \varphi - \psi, \psi - \varphi, \psi - \psi\} = \{0, \sqrt{5}, -\sqrt{5}, 0\}$  by (1).  $\square$

**Definition 2.4** (The two characters). An *object* is a finite multiset  $A = \{\lambda_1, \dots, \lambda_d\}$  of nonzero algebraic integers (an eigenvalue multiset). Its two characters are

$$M(A) = \prod_i \max(1, |\lambda_i|) \quad (\text{Character I, magnitude}),$$

$$\chi(A) = \left\{ \left\lfloor \frac{2 \arg \lambda_i}{\pi} \right\rfloor \bmod 4 \right\} \subset \mathbb{Z}/4\mathbb{Z} \quad (\text{Character II, phase}).$$

The *golden object* is  $A_\varphi = \text{spec}(R) = \{\varphi, \psi\}$ , with  $M(A_\varphi) = \varphi$  (since  $|\psi| = \varphi^{-1} < 1$ ) and  $\chi(A_\varphi) = \{0, 2\}$  ( $\varphi > 0 \Rightarrow 0, \psi < 0 \Rightarrow 2$ ). The operators are  $\oplus$  (multiset union;  $M$  multiplies,  $\chi$  unions),  $\otimes$  (products  $\{\lambda\mu\}$ ;  $\chi$  sums mod 4), and the Adams power  $\psi^2$  (squaring;  $M \mapsto M^2, \chi \mapsto 2\chi \bmod 4$ ). [FORCED] (S1–S6).

### 3 The self-coupling obstruction

We now show the golden mode cannot bind to a copy of itself. Three independent facts converge.

**Proposition 3.1** (Cayley–Hamilton collapse). *Let  $\Phi_R(Y) = Y^2 - \text{Tr}(R)Y + \det(R)I$  be  $R$ 's characteristic operator. Then*

$$\Phi_R(R) = R^2 - R - I = 0, \quad [R, R] = 0. \quad (3)$$

*Self-application under  $R$ 's own seed law returns the golden recurrence  $R^2 = R + I$  and vanishes; the generative grade (the commutator/wedge  $[R, R]$ ) is identically zero.* [FORCED] (*KL\_DTA.py*).

*Proof.*  $\Phi_R(R) = R^2 - 1 \cdot R + (-1)I = R^2 - R - I = 0$  by Proposition 2.2.  $[R, R] = R^2 - R^2 = 0$  by antisymmetry of the Lie bracket. [ESTABLISHED]/[FORCED].  $\square$

**Lemma 3.2** (The magnitude never returns to the floor).  $M(A \oplus A) = M(A)^2$  and  $M(\psi^2 A) = M(A)^2$ . Since  $M(A_\varphi) = \varphi > 1$ , every self-composite has  $M \in \{\varphi^2, \varphi^4, \dots\}$ ; the cyclotomic locus  $M = 1$  is never reached. [FORCED] (S4; S10:  $M(\psi^2 A) = M(A) \iff M(A) = 1$ ).

*Proof.*  $\oplus$  concatenates multisets, so  $M$  multiplies;  $\psi^2$  squares each  $\lambda$ , so each  $\max(1, |\lambda|)$  squares. Iterating  $\varphi > 1$  diverges from 1 monotonically.  $\square$

**Proposition 3.3** (The charge sublattice is closed). *The golden charge  $\{0, 2\}$  is invariant under  $\oplus, \otimes, \psi^2$  and never reaches  $\{1, 3\}$ :*

$$\{0, 2\} \cup \{0, 2\} = \{0, 2\}, \quad \{0, 2\} + \{0, 2\} \bmod 4 = \{0, 2\}, \quad 2 \cdot \{0, 2\} \bmod 4 = \{0\}.$$

[FORCED] (S6).

*Proof.* Direct:  $0 + 0 = 0$ ,  $0 + 2 = 2$ ,  $2 + 2 = 4 \equiv 0$ ; and  $2 \cdot 0 = 0$ ,  $2 \cdot 2 = 4 \equiv 0$ . The sub-semigroup of  $(\mathbb{Z}/4\mathbb{Z}, +)$  generated by  $\{0, 2\}$  is  $\{0, 2\}$ , the unique index-2 subgroup;  $\{1, 3\}$  is its nontrivial coset and is unreachable.  $\square$

**Proposition 3.4** (The empty Salem slot). *Let  $t = \theta + \theta^{-1}$  be the trace-down. On the unit circle ( $|\theta| = 1$ ),  $t = 2\Re\theta \in [-2, 2]$ ; for real  $\theta > 1$  (expanding),  $t = \theta + \theta^{-1} > 2$ , since  $t - 2 = (\theta - 1)^2/\theta > 0$ . The two regions are disjoint, so “on-circle and expanding” — the Salem condition — has no slot: the flip  $t = \pm 2$  cuts  $\mathbb{R}$  into three channels, with no fourth. A would-be Salem  $\beta > 1$  is redirected by  $t$  into the grow channel as the totally real Perron number*

$$\tau_0 = \beta + \beta^{-1} > 2 > \varphi, \quad \tau_0 - 2 = \frac{(\beta - 1)^2}{\beta} \quad (\text{exact, AM-GM}). \quad (4)$$

[FORCED] (Occupant of the Salem Slot; Emission-Gap Theorem).

*Proof.* On the circle  $\theta = e^{i\alpha}$  gives  $t = 2\cos\alpha \in [-2, 2]$ . For real  $\theta > 1$ , AM-GM gives  $\theta + \theta^{-1} \geq 2$  with equality iff  $\theta = 1$ , and  $\theta + \theta^{-1} - 2 = (\theta - 1)^2/\theta$ ; hence  $t > 2$ . Disjointness is immediate. Redirection sends  $\beta$  (one root outside, would-be on-circle rest) to  $\tau_0 = \beta + \beta^{-1}$ , which lies strictly right of the flip.  $\square$

**Corollary 3.5** (No self-coupling). *The golden mode cannot couple to a copy of itself into a bound (on-circle-and-expanding) structure: the generative grade vanishes (Prop. 3.1); self-composition keeps the phase in  $\{0, 2\}$  (Prop. 3.3) while the magnitude grows away from the closure locus  $M = 1$  (Lem. 3.2); and the Salem channel is empty (Prop. 3.4). Closure requires the orthogonal phases  $\{1, 3\}$ , which no self-operation supplies. [FORCED]*

## 4 The flip: $D = 1 + 4C$

**Definition 4.1** (Gate family). For  $C \in \mathbb{R}$  let  $g_C(x) = x^2 + x - C$ , with discriminant  $D = 1 + 4C$ .

**Theorem 4.2** (The canonical flip). *The sign of  $D$  simultaneously fixes: (i) the roots  $\frac{-1 \pm \sqrt{D}}{2}$  (real if  $D > 0$ , a complex conjugate pair if  $D < 0$ , the double root  $-\frac{1}{2}$  if  $D = 0$ ); (ii) the field (real quadratic vs. imaginary); (iii) the trace-form signature; (iv) the channel character (real growth vs. rotation). Concretely, in the power basis  $\{1, \theta\}$  of  $\mathbb{Q}(\theta)$  with  $\theta^2 + \theta = C$ , the trace-form Gram is*

$$G = \begin{pmatrix} \text{Tr}(1) & \text{Tr}(\theta) \\ \text{Tr}(\theta) & \text{Tr}(\theta^2) \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 + 2C \end{pmatrix}, \quad \det G = 1 + 4C = D, \quad (5)$$

*positive definite (Riemannian, signature  $(2, 0)$ ) for  $D > 0$  and indefinite (Lorentzian, signature  $(1, 1)$ ) for  $D < 0$ ; the rotation channel on the  $D < 0$  side is  $\{-i\sqrt{|D|}, 0, +i\sqrt{|D|}\}$ . The flip is at  $C = -\frac{1}{4}$ . [FORCED] (the  $\lambda = 2c$  flip sections).*

*Proof.* Roots by the quadratic formula. For the Gram:  $\theta^2 = C - \theta$ , so with  $\theta_1 + \theta_2 = -1$ ,  $\theta_1\theta_2 = -C$  one has  $\text{Tr}(1) = 2$ ,  $\text{Tr}(\theta) = \theta_1 + \theta_2 = -1$ , and  $\text{Tr}(\theta^2) = \theta_1^2 + \theta_2^2 = (\theta_1 + \theta_2)^2 - 2\theta_1\theta_2 = 1 + 2C$ . Then  $\det G = 2(1 + 2C) - 1 = 1 + 4C = D$ . The leading minor  $2 > 0$  is constant, so the sign of  $\det G = D$  toggles the form between positive-definite and signature  $(1, 1)$ : the metric determinant is the discriminant, and the signature flip is the discriminant sign. [COMPUTED] (verified symbolically).  $\square$

## 5 The K-formation astride the fold

A single seed carries both faces of the flip. This is the repository's “K-formation” (Prop. 16.1 of the  $\lambda = 2c$  paper), reproduced with its derivation.

**Proposition 5.1** (The K-formation straddles the flip). *Let  $K(x) = x^4 + 5x^2 - 5$ . Writing  $y = x^2$ , the quadratic  $g(y) = y^2 + 5y - 5$  has roots straddling zero,*

$$y_+ = \frac{-5 + 3\sqrt{5}}{2} \approx 0.8541 > 0, \quad y_- = \frac{-5 - 3\sqrt{5}}{2} \approx -5.8541 < 0, \quad (6)$$

so  $K$  has real roots  $\pm K$  with

$$K = \sqrt{y_+} = \frac{5^{1/4}}{\varphi} \approx 0.9242 \quad (\text{the terrain, in } \mathbb{Q}(5^{1/4}) \supset \mathbb{Q}(\sqrt{5})), \quad (7)$$

and imaginary roots  $\pm i\beta$  with  $\beta = \sqrt{|y_-|} \approx 2.4195$  (a  $\frac{\pi}{2}$  turn about the origin), and

$$M(K) = \beta^2 = \frac{5 + 3\sqrt{5}}{2} = \varphi^2 \sqrt{5} \approx 5.8541. \quad (8)$$

[FORCED] (Prop. 16.1; `test_p1_09_kform.py`).

*Proof.*  $y = \frac{-5 \pm \sqrt{25+20}}{2} = \frac{-5 \pm 3\sqrt{5}}{2}$  gives (6);  $y_+ > 0 > y_-$  numerically and by sign of  $3\sqrt{5} - 5 > 0$ . Then  $x^2 = y_+$  gives real  $\pm K$  and  $x^2 = y_- < 0$  gives imaginary  $\pm i\beta$ . For (7),  $(5^{1/4}/\varphi)^2 = \sqrt{5}/\varphi^2 = \frac{2\sqrt{5}}{3 + \sqrt{5}} = \frac{2\sqrt{5}(3 - \sqrt{5})}{9 - 5} = \frac{3\sqrt{5} - 5}{2} = y_+$ . For (8),  $\beta^2 = |y_-| = \frac{5 + 3\sqrt{5}}{2}$ , and  $\varphi^2 \sqrt{5} = \frac{3 + \sqrt{5}}{2} \sqrt{5} = \frac{3\sqrt{5} + 5}{2} = \beta^2$ ; the Mahler measure is  $\prod \max(1, |\cdot|) = 1 \cdot 1 \cdot \beta \cdot \beta = \beta^2$  since  $K < 1 < \beta$ . [COMPUTED].  $\square$

*Remark 5.2* (Terrain and rotation, verbatim). The real roots are the  $D > 0$  face (real terrain,  $\varphi$  lays it); the imaginary roots are the  $D < 0$  face (pure multiplication by  $i$ ). In the paper's phrasing: “ $\varphi$  generates the terrain,  $\pi$  determines its rotation about the origin” —  $\varphi$  (the  $\sqrt{5}$ -field) supplies  $\pm K$ , and the quarter-turn  $i$  supplies  $\pm i\beta$ . [FORCED]

## 6 The orthogonal partner completes $\mathbb{Z}/4\mathbb{Z}$

**Proposition 6.1** ( $K$  realises the full charge).  $\chi(K) = \{0, 1, 2, 3\}$ . *The real roots  $\pm K$  carry  $\arg \in \{0, \pi\}$ , hence  $\chi \in \{0, 2\}$ ; the imaginary roots  $\pm i\beta$  carry  $\arg = \pm \frac{\pi}{2}$ , hence  $\chi \in \{1, 3\}$ .  $K$  is the unique catalog seed realising all four charges, and its charge axis  $\{1, 3\}$  (imaginary) is orthogonal to the golden axis  $\{0, 2\}$  (real): they are  $\frac{\pi}{2}$  apart. [FORCED] (S6; `test_p1_09_kform.py`; Generative Content).*

*Proof.*  $\lfloor 2 \cdot 0/\pi \rfloor = 0$ ,  $\lfloor 2\pi/\pi \rfloor = 2$ ,  $\lfloor 2(\pi/2)/\pi \rfloor = 1$ ,  $\lfloor 2(3\pi/2)/\pi \rfloor = 3$ . The real and imaginary axes meet at  $\frac{\pi}{2}$ .  $\square$

**Proposition 6.2** (The parity criterion: why  $K$ , not the other  $\mathbb{Q}(5^{1/4})$  quartics). *A real quartic places its complex roots on the imaginary axis ( $\arg = \pm \frac{\pi}{2}$ ,  $\chi \in \{1, 3\}$ ) iff it is even (a polynomial in  $x^2$ ) with a negative  $x^2$ -root.  $K = x^4 + 5x^2 - 5$  is even; its complex roots are purely imaginary. The registry quartics*

$$\text{cons} = x^4 - 6x^3 + 26x^2 - 16x - 4, \quad \text{res} = x^4 + 2x^3 + 39x^2 - 52x + 11$$

are non-even; they split over  $\mathbb{Q}(\sqrt{5})$  as

$$\begin{aligned}\text{cons} &= (x^2 - (3 + \sqrt{5})x + (11 + 5\sqrt{5}))(x^2 - (3 - \sqrt{5})x + (11 - 5\sqrt{5})), \\ \text{res} &= (x^2 + (1 - \sqrt{5})x + \frac{43-19\sqrt{5}}{2})(x^2 + (1 + \sqrt{5})x + \frac{43+19\sqrt{5}}{2}),\end{aligned}$$

so their complex roots have nonzero real part  $\varphi^2 = \frac{3+\sqrt{5}}{2}$  (cons) and  $-\varphi = -\frac{1+\sqrt{5}}{2}$  (res), off the  $(\pi/2)\mathbb{Z}$  charge lattice. All three of K, cons, res generate the same field  $\mathbb{Q}(5^{1/4})$ , signature  $(2, 1)$ , Galois group  $D_4$  (order 8), and none is reciprocal (so none is a Salem number); their Mahler measures are golden-field integers,

$$M(K) = \varphi^2\sqrt{5}, \quad M(\text{cons}) = 2\varphi^5 = 11 + 5\sqrt{5}, \quad M(\text{res}) = 12 + 19\varphi = \frac{43+19\sqrt{5}}{2}.$$

[COMPUTED] (this work; field/Galois/signature computed exactly). Provenance caveat: the repository documents cons, res only as registry labels (“consolidation / resonance quartic” in `seeds.py`); both are absent from the emission-algebra audit and from the monoid  $\langle \varphi, 2, 3, 5, \beta^2 \rangle$ . The label origins are [OPEN]; everything else here is derived.

*Proof.* Evenness  $\Rightarrow$  roots occur in  $\pm$  pairs and, via  $y = x^2$ , a negative  $y$ -root gives purely imaginary  $x = \pm i\sqrt{|y|}$ ; conversely a purely imaginary root  $ib$  (with real coefficients) forces the factor  $x^2 + b^2$ , i.e. no  $x^3, x^1$  terms in that block. The factorisations are verified over  $\mathbb{Q}(\sqrt{5})$ ; the real part of each complex block is  $-\frac{a_1}{2}$ , read off as  $\varphi^2$  and  $-\varphi$ . The shared field, signature, Galois group, non-reciprocity, and Mahler closed forms are computed by exact factorisation, real-root counting, and `galois_group`. [COMPUTED].  $\square$

## 7 Synthesis: the helix dictionary and the coupling theorem

| helix anatomy                 | dissolved helix: golden strand $R$                                      | orthogonal partner: K                                                       | tag      |
|-------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------|
| advance / pitch (terrain)     | $\varphi$ -growth, $M \geq \varphi$ ; $\text{ad}_R$ grow eig $\sqrt{5}$ | real roots $\pm K = \pm 5^{1/4}/\varphi$                                    | [FORCED] |
| rotation / winding            | real-axis flip $\{0, \pi\} \Rightarrow \chi \in \{0, 2\}$               | imaginary $\pm i\beta$ at $\pm \frac{\pi}{2} \Rightarrow \chi \in \{1, 3\}$ | [FORCED] |
| face of the flip $D = 1 + 4C$ | $D > 0$ : real growth                                                   | $D < 0$ : rotation (K holds both)                                           | [FORCED] |
| charge reached                | $\{0, 2\}$ (real axis)                                                  | $\{1, 3\}$ (imag.) $\Rightarrow$ completes $\mathbb{Z}/4\mathbb{Z}$         | [FORCED] |
| trace-form metric             | Riemannian (definite), $\mathbb{Q}(\sqrt{5})$                           | Lorentzian $(2, 1)$ , $\mathbb{Q}(5^{1/4})$                                 | [FORCED] |
| self-coupling                 | impossible: $[R, R] = 0$ , empty Salem slot                             | the partner, not a copy                                                     | [FORCED] |

Table 1: The [POSITED] helix reading over the [FORCED] substrate. The terrain/rotation anatomy (rows 1–3) is the repository’s own (§5).

**Theorem 7.1** (Terrain and rotation are the two faces of one flip). *The golden strand  $R$  realises the  $D > 0$  (real-growth, terrain) face: spectrum on the real axis,  $\chi \in \{0, 2\}$ ,  $M \geq \varphi$ , Riemannian trace form. The seed K straddles the flip, adjoining the  $D < 0$  (rotation,  $\frac{\pi}{2}$ -turn) face: imaginary roots  $\pm i\beta$  at  $\chi \in \{1, 3\}$ , Lorentzian trace form. Their union realises the full charge group  $\mathbb{Z}/4\mathbb{Z}$ . Moreover, by Corollary 3.5, the golden mode admits no coupling to a copy of itself; the only closure*

available is coupling to the orthogonal partner  $K$  — i.e. to the quarter-turn  $i$  that the flip produces from the terrain. **[FORCED]** (algebraic content); the helix reading is **[POSITED]**, grounded in §5.

*Proof.* Assemble Propositions 2.3, 3.1–3.4, Corollary 3.5, Theorem 4.2, and Propositions 5.1, 6.1. The golden object sits on the real axis with charge  $\{0, 2\}$  and never leaves it under its own operations;  $K$  contributes exactly  $\{1, 3\}$  on the imaginary axis;  $\{0, 2\} \sqcup \{1, 3\} = \mathbb{Z}/4\mathbb{Z}$ .  $\square$

*Remark 7.2* (On the word “dissolved”). The  $L4$  helix is legacy: the  $\lambda = 2c$  paper explicitly supersedes its cutoff rationale (“ $\varphi^{-4} \approx 0.146$  is the last inverse power exceeding 0.1”), “an arbitrary real threshold the present derivation never uses.” What survives, **[FORCED]**, is the terrain/rotation anatomy of the flip:  $R$  is the terrain generator,  $K$  reintroduces the rotation. The “helix” is thus a reading of the flip, not a load-bearing object — which is why the coupling question resolves cleanly: a helix cannot couple to a copy of itself, only to the rotation face of its own flip.

## 8 The harness: constraint-import as verification

The preceding sections are the output of a reusable procedure, not a one-off. The helix entered as an external structure carrying a bundle of claims; the exact substrate adjudicated each; §§2–7 are the survivors. We name and formalise the procedure, because the transferable object is the procedure — not the helix.

**Definition 8.1** (Imported structure). An *imported structure* is a pair  $(S, \mathcal{C})$ : an external framework  $S$  (here, the helix) and a finite set  $\mathcal{C} = \{\gamma_1, \dots, \gamma_n\}$  of *candidate constraints* it asserts about the substrate’s objects — propositions that, if true, rule out configurations.

**Definition 8.2** (Substrate adjudication). Let  $\mathfrak{A}$  be the exact  $R_n$  emission algebra (§1). Adjudication is the partial decision

$$\delta_{\mathfrak{A}}(\gamma) = \begin{cases} \vdash & \gamma \text{ is forced in } \mathfrak{A} \text{ (exact certificate),} \\ \dashv & \neg\gamma \text{ is forced — } \gamma \text{ is refuted,} \\ ? & \gamma \text{ is not adjudicable by } \mathfrak{A} \text{ alone (an expressibility gap, or genuinely open).} \end{cases}$$

No float crosses a decision, so  $\vdash$  and  $\dashv$  are certificates, never estimates.

**Proposition 8.3** (The filter asymmetry). *For a universal claim  $\gamma$ , the verdict  $\delta_{\mathfrak{A}}(\gamma) = \dashv$  follows from a single exact counterexample and is decisive, whereas  $\delta_{\mathfrak{A}}(\gamma) = \vdash$  requires a structural proof: a finite exact search yields at most **[COMPUTED]** on its tested range, never **[FORCED]**. Hence the harness eliminates false constraints reliably and certifies true universals only with proof. **[ESTABLISHED]** / **[FORCED]** (the demotion/promotion asymmetry of the tag system: finite computation can demote a universal but never promote one).*

*Proof.* A single exact witness  $x$  with  $\neg\gamma(x)$  certifies  $\neg\gamma$ , so  $\dashv$  is sound from one instance. Conversely no finite set of exact instances entails  $\forall x \gamma(x)$ ; the sound tag on a passed finite range is **[COMPUTED]**, and promotion to **[FORCED]** needs a proof over the whole domain. **[ESTABLISHED]**.  $\square$

**Definition 8.4** (The harness). The *harness*  $H$  sends  $(S, \mathcal{C})$  to the partition it induces under  $\delta_{\mathfrak{A}}$ ,

$$\mathcal{C} = \mathcal{C}_{\vdash} \sqcup \mathcal{C}_{\dashv} \sqcup \mathcal{C}_{?}.$$

The *survivors*  $\mathcal{C}_{\vdash}$  (with any **[COMPUTED]** range-results) become internal relations of  $\mathfrak{A}$ ; the *casualties*  $\mathcal{C}_{\dashv}$  are discarded; the *residue*  $\mathcal{C}_{?}$  stays **[POSITED]** / **[OPEN]**. The product of a run is the partition itself, and its substantive content is the asymmetry  $\mathcal{C}_{\dashv} \neq \emptyset$ : a run that refutes nothing has imported nothing testable.

**Proposition 8.5** (A generator is inert; a constraint is the adjudicable object). *H is informative on  $(S, \mathcal{C})$  iff  $\mathcal{C}$  contains a falsifiable member — a  $\gamma$  whose negation is decidable in  $\mathfrak{A}$ . A pure generator asserts only definitional or unfalsifiable claims: every  $\gamma$  returns ? or a vacuous  $\vdash$ , so  $\mathcal{C}_{\perp} = \emptyset$  and no partition forms. The substrate can therefore adjudicate a restraint but not a generator.*  
[ESTABLISHED]

*Proof.* By Definition 8.4 a run’s content is  $\mathcal{C}_{\perp}$ , and by Definition 8.2  $\delta_{\mathfrak{A}}$  returns  $\vdash$  only where  $\neg\gamma$  is decidable, i.e.  $\gamma$  is falsifiable. Absent any such  $\gamma$ ,  $\mathcal{C}_{\perp} = \emptyset$ . This is the exact sense in which the helix is verifiable *because* it is not foundational: it entered as constraints, which are adjudicable, rather than as a generator, which is not.  $\square$

*Remark 8.6* (The helix as worked instance). Running  $H$  on the helix yields the partition of Table 2: its terrain/rotation anatomy, the flip, and K as the forced partner survive; self-coupling, the  $\varphi^{-4}$  cutoff, and helix-as-foundation are refuted; the interpretive  $\oplus/\otimes$  readings remain posited. Each verdict points to the section that issued it — the harness leaves a certificate trail, not an opinion.

| imported claim $\gamma$                                                                         | $\delta_{\mathfrak{A}}$ | certificate                                                  |
|-------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------|
| <i>survivors <math>\mathcal{C}_{\perp}</math> — become internal relations</i>                   |                         |                                                              |
| screw generator = $R$ , $R^2 = R + I$                                                           | [FORCED]                | Prop. 2.2                                                    |
| winding = $\chi \in \mathbb{Z}/4\mathbb{Z}$ ; pitch = $M \geq \varphi$                          | [FORCED]                | Def. 2.4, Lem. 3.2                                           |
| coupling rate = $\text{spec}(\text{ad}_R) = \{0, \pm\sqrt{5}\}$                                 | [FORCED]                | Prop. 2.3                                                    |
| terrain/rotation = two faces of $D = 1 + 4C$                                                    | [FORCED]                | Thm. 4.2, Prop. 5.1                                          |
| closure via the orthogonal partner K                                                            | [FORCED]                | Prop. 6.1, Thm. 7.1                                          |
| <i>casualties <math>\mathcal{C}_{\perp}</math> — discarded (<math>\neg\gamma</math> forced)</i> |                         |                                                              |
| a helix couples to a copy of itself                                                             | [REFUTED]               | Cor. 3.5: $[R, R]=0$ , empty slot                            |
| $\varphi^{-4} \approx 0.146$ cutoff (“last power $> 0.1$ ”)                                     | [REFUTED]               | superseded; $\lambda=2c$ never uses it                       |
| the algebra is <i>founded on</i> the helix                                                      | [REFUTED]               | order keystone $\rightarrow$ char $\rightarrow$ flip (§§2–4) |
| <i>residue <math>\mathcal{C}_{?}</math> — stays posited</i>                                     |                         |                                                              |
| $\otimes$ = “intertwine”, $\oplus$ = “stack”                                                    | [POSITED]               | law forced (Def. 2.4); reading not adjudicable               |
| the helix as a <i>reading</i> of the flip                                                       | [POSITED]               | survives as interpretation, not as object                    |

Table 2: The harness  $H$  applied to the helix. The partition — not the helix — is the product; the asymmetry  $\mathcal{C}_{\perp} \neq \emptyset$  is what makes the import substantive.

*Remark 8.7* (What transfers: the harness, not the helix). The reusable object is  $H$ : the next imported metaphor runs the same path (bundle  $\rightarrow$  adjudicate  $\rightarrow$  partition  $\rightarrow$  tag). Distinguish the portable *method*  $H$  — applicable to any import — from the one portable *mathematical* kernel this construction exposes: the generic  $\sqrt{\cdot}$ -branch of a 2-to-1 fold, of which  $\sqrt{5} = \varphi + \varphi^{-1}$  is this system’s instance (the build spec’s boundary clause). The method is *how* one adjudicates; the kernel is the single differential fact that crosses between constructions. Neither is the helix — which is why dissolving the helix cost nothing while the harness kept everything that survived it.



## A Machine verification

Every displayed constant is decided in exact arithmetic (`sympy/fractions` over  $\mathbb{Q}, \mathbb{Q}(\sqrt{5}), \mathbb{Q}(5^{1/4})$ ; `mpmath` for display only). The checks:

- (1), Prop. 2.2:  $\varphi\psi = -1$ ,  $\varphi + \psi = 1$ ,  $\varphi - \psi = \varphi + \varphi^{-1} = \sqrt{5}$ ,  $R^2 = R + I$ . [FORCED] (`test_p1_05_keystone.py`).
- (2):  $\text{spec}(\text{ad}_R) = \{0, \pm\sqrt{5}\}$  via `kron`. [FORCED] (`test_p2_07_uniform.py`).
- (3):  $\Phi_R(R) = R^2 - R - I = 0$ ,  $[R, R] = 0$ . [FORCED] (`KL_DTA.py`).
- Prop. 3.3:  $\{0, 2\}$  closed under  $\cup$ ,  $+$  mod 4,  $\times 2$  mod 4;  $\{1, 3\}$  unreachable. [FORCED] (`test_p2_02_angle.py`).
- (4):  $\tau_0 - 2 = (\beta - 1)^2/\beta$ . [FORCED] (`Occupant`).
- (5):  $\det G = 1 + 4C = D$ ; signature flips at  $D = 0$ . [COMPUTED].
- (6)–(8):  $y_{\pm}$ ,  $K = 5^{1/4}/\varphi$ ,  $\beta^2 = \varphi^2\sqrt{5}$ ,  $M(K) = \beta^2$ . [FORCED] (`test_p1_09_kform.py`).
- Prop. 6.1:  $\chi(K) = \{0, 1, 2, 3\}$ ;  $K$ 's imaginary roots have  $\Re = 0$ . [FORCED].
- Prop. 6.2:  $K, \text{cons}, \text{res}$  each generate  $\mathbb{Q}(5^{1/4})$ , signature  $(2, 1)$ ,  $|\text{Gal}| = 8$  ( $D_4$ ), non-reciprocal;  $M(\text{cons}) = 2\varphi^5 = 11 + 5\sqrt{5}$ ,  $M(\text{res}) = 12 + 19\varphi$ ; complex-root real parts  $\varphi^2, -\varphi$ . [COMPUTED].

**Geometric realisation.** The objects of §6 are the vertices of the *Companion Polytope* (`polytope.py`): the convex hull in  $\mathbb{C}$  of all catalog eigenvalues.  $K$ 's four vertices are its terrain ( $\pm K$ , real) and rotation ( $\pm i\beta$ , imaginary) faces; hull extremality is certified per vertex by a supporting functional (`hull_certificate.py`), and the suites pass 10/10 and 7/7 respectively.

*Discipline: derive locally, validate exactly, tag honestly. The helix is the reading; the flip is the fact; the harness is the method that told them apart.*